**Additional Information and FAQ for Work Assignment 1**

**Frequency Evaluation: Mandatory Work Assignment #1**

**3S – Secure Streaming Solution: PART I**

# Design, Implementation, and Experimental Evaluation of a Security Solution for Real-Time Media Streaming using Flexible Cryptographic Configuration

## Complementary information for Work-Assignment 1

1. Submission Deadline
2. Required Submission Materials
3. Submission Procedure
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## 1. Submission Deadline

- Submission deadline date/time: THU, 29 October 2026, 23:59:59
  Note: you must manage your agenda for the established deadline

## 2. Required Submission Materials

For the work submission, students must submit the following materials:

### 2.1 Source Code Project

A complete ZIP archive containing the source code and the full project structure required to verify, compile, and execute the implementation.

The archive must be named according to the following conventions:

- **Group submission:** NNNNN-MMMM-WA1.zip, where NNNNN and MMMM are the student numbers of the two group members.
- **Individual submission:** NNNNN-WA1.zip, where NNNNN is the student number of the individual student.

The ZIP archive must include a README file containing all the information required to compile and run the implementation from the command line. The instructions must support an initial, rapid verification of the submitted solution.

The README may also include example commands, test scenarios, and expected outputs to facilitate the verification that the implementation behaves as specified.

### 2.2 Technical Report

Students must submit a technical report in PDF format (as a typical paper), approximately **4-6 pages** in length, describing the designed security solution

The report must document the architecture and specifications of the implemented solution, including the main components, their responsibilities, the communication protocols, and the security mechanisms involved. It should also describe the experimental evaluation, the observed results, and the evidence supporting the correctness of the implementation.

The PDF file must be named according to the following conventions:

- **Group submission:** NNNNN-MMMM-WA1-Report.pdf, where NNNNN and MMMM are the student numbers of the two group members.
- **Individual submission:** NNNNN-WA1-Report.pdf, where NNNNN is the student number of the individual student.

The report should follow a recognized scientific-paper format and use an appropriate academic template. For example, students may consult the Springer information for authors and editors.

Possible structure (as an indicative reference)

Title

Author or Authors

Abstract and keywords

1. Introduction
2. System model and architrectural design
3. Threat model definition and security countermeasures
4. Specification of the Secure Streaming Protocol
5. Cryptographic configuration files
6. Stream Proxy design and implementation options
7. Server Stream design and implementation options
8. Validation
   - 8.1 Conducted tests and experimental validation testbench
   - 8.2 Experimental results from experimental observations
9. Conclusions

The quality of the report will be considered in the evaluation, together with the completeness and quality of the designed solution, implementation and validation, in compliance with the assignment requirements, and the quality of the experimental evaluation and evidence of correctness.

## 3. Submission Procedure

By **26 October**, students will receive a message through CLIP containing a submission form. The form must be used to submit the materials specified above. You must be ready with all the materials for the submission

The following rules apply:

- The form must be completed and submitted by the submission deadline.
- For group assignments, one group member may submit the form on behalf of the entire group.
- For individual assignments, the student must submit the form personally.

The form will provide sections for uploading:

- The source code archive (ZIP file), including the README file referred in 2.1 and explained in 4.5
- The technical report (PDF file): we expect that good reporting can be written on 4 to 6 pages max, with a format inspired in the above indicative reference in 2.2
- Additionally, the form will include sections with questions intended to clarify and characterize the submitted solution, the objectives achieved, and the experimental results obtained.

Students are responsible for ensuring that both files (source code archive and technical report PDF file) are correctly prepared, named, and uploaded in compliance with the deadline.

## 4. Frequently Asked Questions and Additional Information

### 4.1 Is the work assignment mandatory for course attendance approval and assessment?

Yes. Completing this work assignment is mandatory to obtain attendance approval. Please refer to the course assessment methods and regulations published on CLIP for further details.

### 4.2 Can the assignment be completed individually?

Yes. Students may complete the assignment individually or in pairs.

The estimated workload, expressed in person-hours, and the scope of the required challenges are based on a group of two students. Consequently, students completing the assignment individually should expect a substantially higher individual workload.

Individual submissions will be assessed individually. Each student must demonstrate their understanding of the implemented solution in accordance with the applicable assessment criteria. These may include an individual written test, a demonstration of the implementation, and, if required, an in-person oral examination.

**4.3 Can I use any programming language to implement the solution, or is Java mandatory?**

The assignment may be implemented in any programming language, provided that the resulting solution satisfies all the specified functional and security requirements, which represent the minimum requirements for compliance.

However, the initial reference materials provided for the assignment are written in Java. The presentations, clarifications, code examples, and discussions in the lab sessions may also introduce implementation challenges based on these Java examples. Completing these lab challenges is not mandatory for continuous assessment. Nevertheless, they provide a useful starting point for understanding and addressing the requirements of the mandatory assignments, including Work Assignment 1.

**4.4 Is a specific IDE or development environment mandatory?**

No. Students may use any suitable integrated development environment (IDE), development environment, or toolchain. Students are encouraged to use programming environments and tools with which they are already familiar to maximize their productivity and development efficiency. Examples include IntelliJ IDEA, Eclipse, Visual Studio, or a shell-based environment using command-line tools for compilation, execution, testing, and experimental evaluation.

For Java implementations, students may organize their projects as Maven projects or use another appropriate Java project structure. Implementations in other programming languages may use the corresponding build systems and development tools.

The submitted ZIP archive may preserve the project structure used during development and must include all files required to inspect, compile, and execute the solution. Any external dependencies must be clearly documented.

**4.5 How should the solution be prepared for evaluation?**

As explained above, the project must include a README file providing clear instructions on how to compile and run the solution.

The README file must explain:

- **How to compile the code:** Provide shell commands that can be executed from the project root directory to compile the Secure Proxy and Secure Stream Server, using command-line tools such as javac or Maven (mvn), as applicable.
- **How to run the code:** Provide shell commands that can be executed from the project root directory to run the Secure Proxy and Secure Stream Server, using Java (java) or Maven (mvn), as applicable. Specify the required cryptographic configuration files and indicate where these files must be placed.
- **How to run the distributed system:** Ensure that the instructions explain how to run the Secure Proxy and Secure Stream Server as separate processes on different physical machines.

The instructions must be sufficiently detailed to allow the submitted solution to be compiled, configured, and executed in the evaluation environment.

**4.6 Can I use Docker for my implementation?**

Yes. Using Docker is encouraged and may be viewed positively during the evaluation. Students may provide Docker containers for the Proxy and the Stream Server. The README file must include clear instructions on how to build and run the provided containers.

The containers must support external configuration files. This allows the configuration of the running containers to be modified without rebuilding the Docker images, particularly when the configuration includes cryptographic settings. Cryptographic configuration files should therefore be provided externally rather than embedded in the container images.

### 4.7 How will the submitted work be evaluated?

The assignment will be evaluated in accordance with the principles, methods, and rules defined for the course and published on CLIP.

The following aspects will be considered during the evaluation:

- **Submission compliance.** Only submissions received by the deadline and prepared in accordance with the specified submission requirements will be considered for evaluation.
- **Timely submission of the evaluation form.** Students must answer all questions in the submission form accurately and ensure that their answers are consistent with the submitted implementation, the objectives achieved, and the experimental results reported. Any inconsistencies between the submitted answers and the actual implementation will be considered an important evaluation factor.
- **Implementation and experimental evaluation.** The submitted solution will be assessed in terms of its completeness, quality, and compliance with the assignment requirements. Students must provide experimental evidence supporting the functionality and security properties of their solution. The implementation will also be subject to correctness checks.
- **Individual assessment.** Each student must complete an individual, in-person written test covering aspects of the developed solution. The test will assess the student's understanding of the design decisions, implementation details, security mechanisms and their rationale, as well as the experimental results obtained. This requirement applies to all students, including those who completed the assignment as part of a group.

The following indicative weighting applies, subject to the official evaluation rules published on CLIP:

- **75% — Implementation correctness and completeness, and experimental evaluation.** This component assesses the implemented solution and the experimental results obtained, according to the specific criteria defined in the assignment specification.
- **25% — Individual written test.** This component assesses the student's knowledge and understanding of the work developed and submitted.

The final grade will be awarded on a **0–20 scale**.

Depending on the evaluation process, students or groups may also be required to demonstrate their work and participate in an oral discussion. These activities may take place during the semester's final evaluation period, with the dates communicated separately.

### 4.8 Which technologies may be used to implement secure communication among the Proxy, Streaming Server, and Authentication Server?

Students must design and specify a complete security solution that addresses the requirements and objectives defined in the assignment specification. This includes identifying the relevant components and their interfaces, defining the communication protocols, and specifying the mechanisms required to provide the required security guaranIlaR3ance gt3DcRrlRrltl2acl

The initial reference implementation of the Streaming Server and Proxy uses TCP sockets to support HTTP-based interactions. Students should take this implementation into account when deciding how to structure and implement their solutions. Regardless of the technology selected, the resulting system must satisfy all the security requirements specified in the assignment.

**4.9 Is the use of TLS or HTTPS recommended or mandatory?**

The use of TLS, TLS-enabled sockets, or HTTPS-based REST interfaces is **neither mandatory nor the primary focus of this assignment**.

The main objective is to design, specify, implement, and experimentally evaluate the security protocols and mechanisms required to satisfy the assignment's security requirements. Using TLS or HTTPS is not a prerequisite for achieving this objective, nor does it automatically guarantee compliance with the assignment requirements.

If TLS or HTTPS is used as the underlying transport-security mechanism for the required protocols, students must still design and implement an appropriate application-level security solution that addresses all the requirements specified in the assignment, including any security properties not provided by the underlying transport-security mechanism.

Students should also be aware that using TLS or HTTPS may introduce additional implementation complexity. They must therefore justify their design choices and clearly explain how the security guarantees provided by the chosen communication technology relate to and complement those provided by their own protocols in satisfying the assignment requirements.

**4.10 What should I do if I have further questions about the work assignment, submission, or evaluation?**

Please direct your questions to the instructor during lab sessions or designated contact hours. Avoid relying on email for urgent questions that require immediate answers, particularly close to submission deadlines.

If questions arise that are relevant to all students, the corresponding clarifications may be incorporated into this document through subsequent updates.